

Cloud Custodian has a number of benefits:

- Maintains real-time compliance
- Governs cost management
- It's use of readymade workflows, queries and policies makes life easy
- Can be run locally, on a remote VM or serverless
- Supports managing of most public cloud vendors like AWS, Azure and GCP

Cloud Custodian is a Python based application and supports Python 2 and 3 versions. You can run a two line command that will install Cloud custodian in minutes.

To install it on Windows, use the following code:

```
$ python3 -m venv custodian
$ ./custodian/Scripts/activate
(custodian) $ pip install c7n
==> Install AWS package
(custodian) $ pip install c7n_azure
==> Install Azure package
(custodian) $ pip install c7n_gcp
==> Install GCP Package
```

To install it on Linux server, give the following commands:

```
$ python3 -m venv custodian
$ source custodian/bin/activate
(custodian) $ pip install c7n
==> Install AWS package
(custodian) $ pip install c7n_azure
==> Install Azure package
(custodian) $ pip install c7n_gcp
==> Install GCP Package
```

Cloud Custodian is ideal to monitor well tagged resources only. If you are not managing tags well, you will be bottlenecked if you use Custodian. By default, Cloud Custodian generates CloudWatch metrics on each policy for the number of resources that match the tag key value, the time to retrieve and filter the resources, and the time to execute actions. Custodian uses the YAML format to keep its policies.

To get the metrics output, type:

```
$ custodian run -s <output_directory>
--metrics aws <policyfile>.yaml
```

To send the metrics to a new name space in CloudWatch, give the following command:

```
$ custodian run -s <output_directory>
--metrics aws:///namespace=foo
```

To send metrics to a specific region, use the following command:

```
$ custodian run -s <output_directory>
--metrics aws:///region=ap-south-1
```

Netdata

This is another open source tool to monitor IT resources in real-time. Interactive visualisation is the advantage of using Netdata. It is an agent based tool that collects thousands of metrics from servers. Netdata has 100+ plugins pre-built to offer more suitable tools for monitoring multiple types of workloads.

Once the agents collect the data, the data/metrics are stored in a tiny internal database. After that, Netdata checks the health and triggers alarms if needed. Netdata can be easily integrated with other monitoring tools as well, if needed. The visual dashboard is optimised for visual anomaly detection, low latency and high resolution, which makes the tool very attractive. Refer to Figure 8 for the sample dashboard.

The GitHub repository for Netdata's open source monitoring agent is powered by nearly 300 contributors. You can use Netdata to monitor physical and virtual servers, containers, IoT/edge devices, Linux, FreeBSD, macOS, Kubernetes, Docker and all their derivatives.

The Netdata site provides an automatic one-line installation script for all Linux distributions, making your life easy during installation. The one line given below will install Netdata on your server. It's magic, isn't it?

```
bash <(curl -Ss https://my-netdata.io/kickstart.sh)
```

Apart from this, Netdata offers unique installation methods for the public cloud, Docker, Kubernetes, FreeBSD, MacOS, etc, along with manual installation. Netdata maintains two releases — Nightly and Stable. We recommend you to use Stable unless there is any requirement to use the Nightly version.

Netdata's internal database uses RAM to store recent metrics, ensuring dashboards and API queries are extremely responsive, while 'spilling' historical metrics to disk. This configuration keeps RAM usage low while allowing for long-term, on-disk metrics storage.

Netdata collects data automatically when you follow standard installation procedures. The key difference in using Netdata over other tools is that its internal plugins run within the Netdata daemon, while external plugins are independent processes that send metrics to Netdata over pipes. There are also plugin orchestrators, which are external plugins with one or more data collection modules.

Its features include:

- Auto-detection of hundreds of metrics
- Real-time monitoring
- No time consuming configuration
- Good visual dashboard
- Hundreds of pre-built plugins
- Low-latency visualisations to detect anomalies
- Lightweight database 

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